In [ ]:
```
# EXPERIMENT - 6
## DATA VISUALIZATION WITH SEABORN

PIN Number:`241UDS0103`

### Aim
To visualize and analyze a library-visits dataset using Seaborn plots,
including scatter plots, KDE plots, histograms, count plots, bar plots,
boxplots, strip plots, joint plots, and heatmaps.
```

In [ ]:
```
## Step 1 – Import the Required Libraries & Dataset
```

In [1]:
```
# Import the required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the dataset
df_241UDS0103 = pd.read_csv("Experiment_6_241UDS0103_Library_Visits.csv")

# Display the dataset
df_241UDS0103.head()
```

Out[1]:

	city	date	library_visits	state
0	Visakhapatnam	2022-01-15	28500	Coastal Andhra
1	Vijayawada	2022-01-15	31800	Rayalaseema
2	Guntur	2022-01-15	35200	North Andhra
3	Tirupati	2022-01-15	39000	Central Andhra
4	Kakinada	2022-01-15	42500	South Andhra

In [ ]:
```
## Step 2 – Info of the Dataset
```

In [2]:
```
# Info of the dataset
df_241UDS0103.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 80 entries, 0 to 79
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   city            80 non-null     str
 1   date            80 non-null     str
 2   library_visits  80 non-null     int64
 3   state           80 non-null     str
dtypes: int64(1), str(3)
memory usage: 2.6 KB
```

In [ ]:
```
## Step 3 — Region-wise Data Points
```

In [3]:
```
# Region-wise data points in the dataframe
df_241UDS0103.value_counts("state")
```

Out[3]:
```
state
Coastal Andhra    16
Rayalaseema       16
North Andhra      16
Central Andhra    16
South Andhra      16
Name: count, dtype: int64
```

In [ ]:
```
## Step 4 — Convert Date Column to Datetime
```

In [4]:
```
# Convert date column to datetime format
df_241UDS0103["date"] = pd.to_datetime(df_241UDS0103["date"])

# Create year column for heatmap analysis
df_241UDS0103["year_241UDS0103"] = df_241UDS0103["date"].dt.year

df_241UDS0103.head()
```

Out[4]:

	city	date	library_visits	state	year_241UDS0103
0	Visakhapatnam	2022-01-15	28500	Coastal Andhra	2022
1	Vijayawada	2022-01-15	31800	Rayalaseema	2022
2	Guntur	2022-01-15	35200	North Andhra	2022
3	Tirupati	2022-01-15	39000	Central Andhra	2022
4	Kakinada	2022-01-15	42500	South Andhra	2022

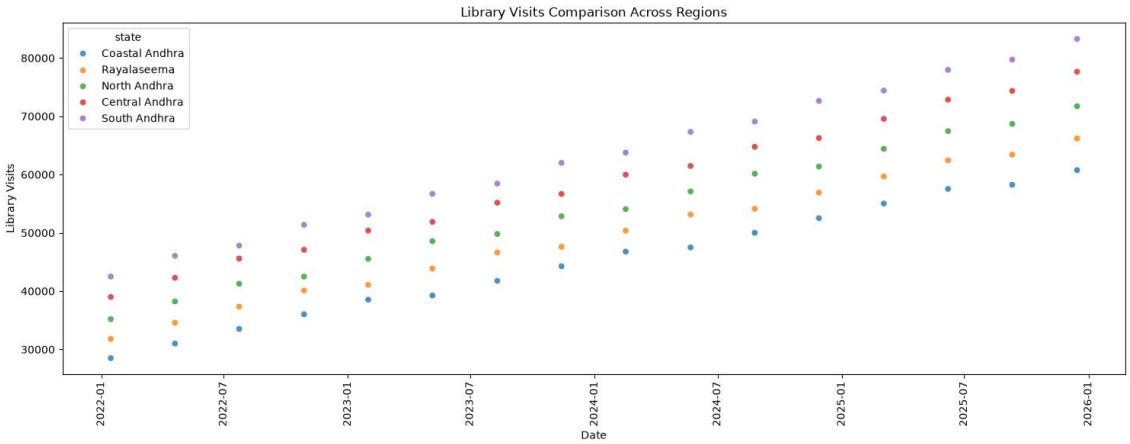
In [ ]:
```
## Step 5 — Visualize Data Using Various Plots
```

In [5]:

```
plt.figure(figsize=(15, 6))

sns.scatterplot(
    x="date",
    y="library_visits",
    hue="state",
    data=df_241UDS0103,
    alpha=0.8
)

plt.title("Library Visits Comparison Across Regions")
plt.xticks(rotation=90)
plt.xlabel("Date")
plt.ylabel("Library Visits")
plt.tight_layout()
plt.show()
```



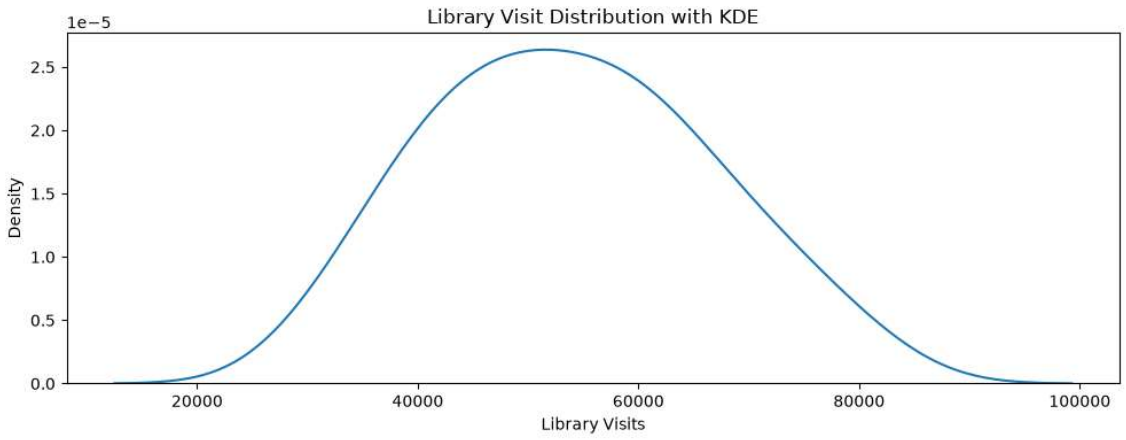
In [ ]:

```
## Step 6 — Histograms & KDE
```

In [6]:
```
plt.figure(figsize=(10, 4))

sns.kdeplot(
    data=df_241UDS0103,
    x="library_visits"
)

plt.title("Library Visit Distribution with KDE")
plt.xlabel("Library Visits")
plt.ylabel("Density")
plt.tight_layout()
plt.show()
```

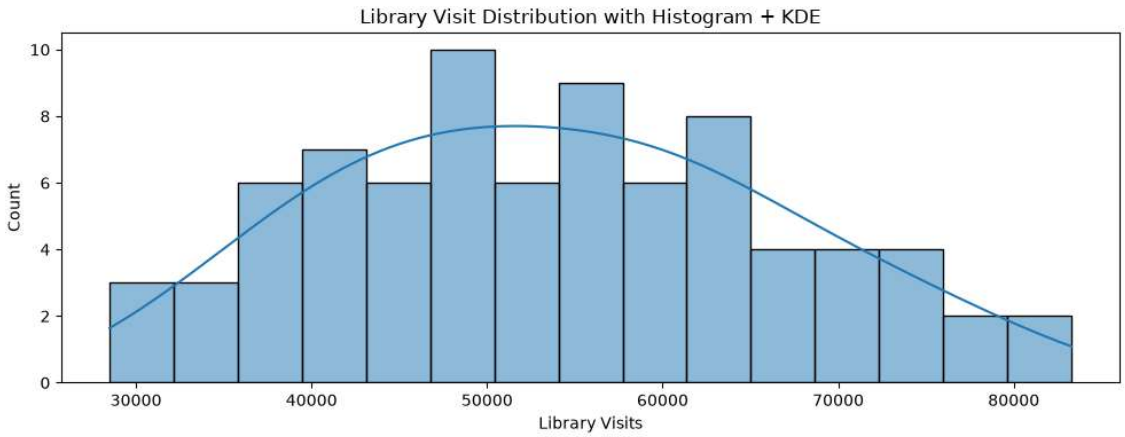


In [ ]:
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## Step 7 — Histogram + KDE
```

In [7]:
```
plt.figure(figsize=(10, 4))

sns.histplot(
    data=df_241UDS0103,
    x="library_visits",
    bins=15,
    kde=True
)

plt.title("Library Visit Distribution with Histogram + KDE")
plt.xlabel("Library Visits")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```

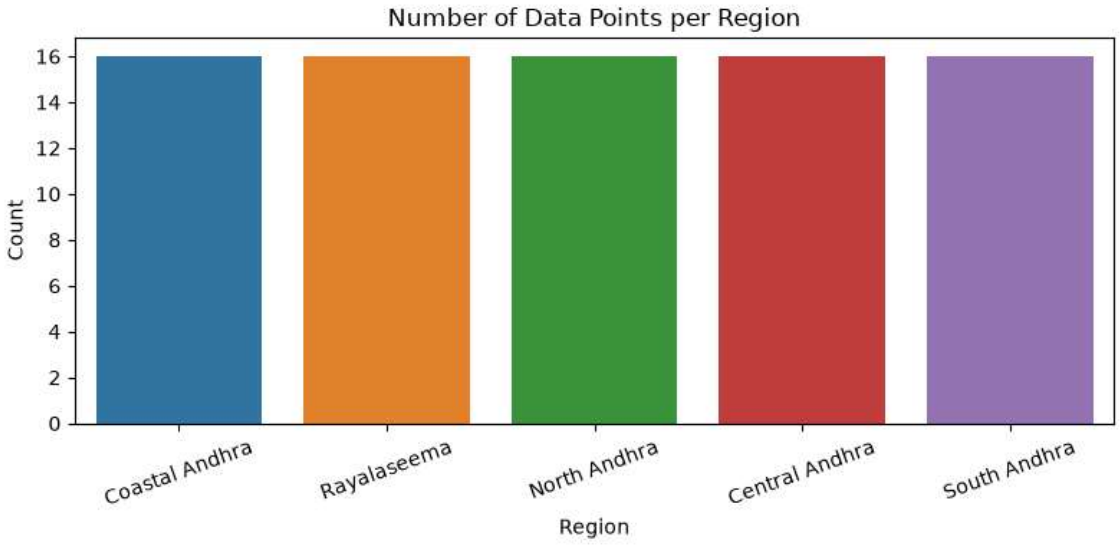


In [ ]:
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## Step 8 — Count Plot
```

In [8]:
```
plt.figure(figsize=(8, 4))

sns.countplot(
    x="state",
    data=df_241UDS0103,
    hue="state",
    legend=False
)

plt.title("Number of Data Points per Region")
plt.xlabel("Region")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

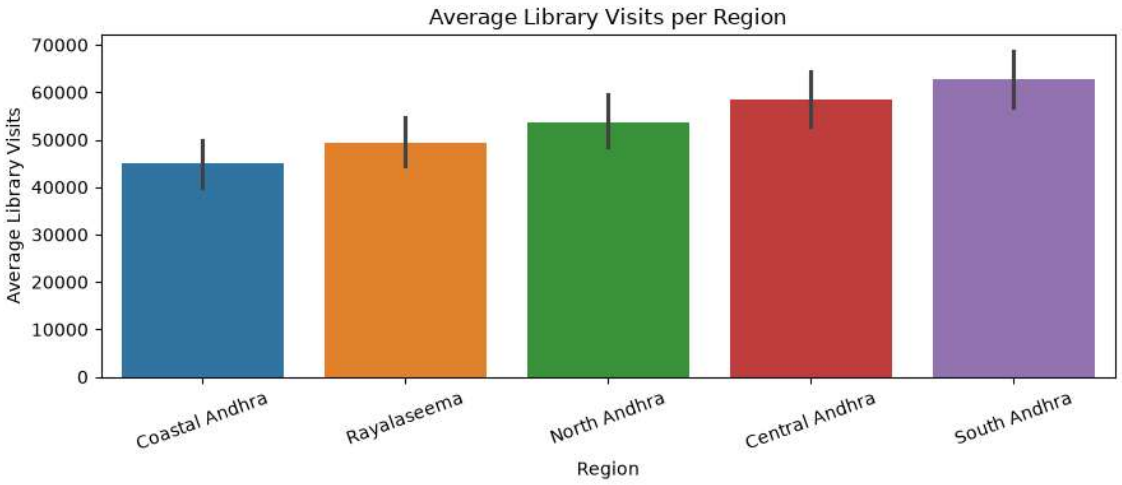


In [ ]:
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## Step 9 – Bar Plot
```

In [9]:
```
plt.figure(figsize=(9, 4))

sns.barplot(
    x="state",
    y="library_visits",
    data=df_241UDS0103,
    hue="state",
    estimator=np.mean,
    legend=False
)

plt.title("Average Library Visits per Region")
plt.xlabel("Region")
plt.ylabel("Average Library Visits")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

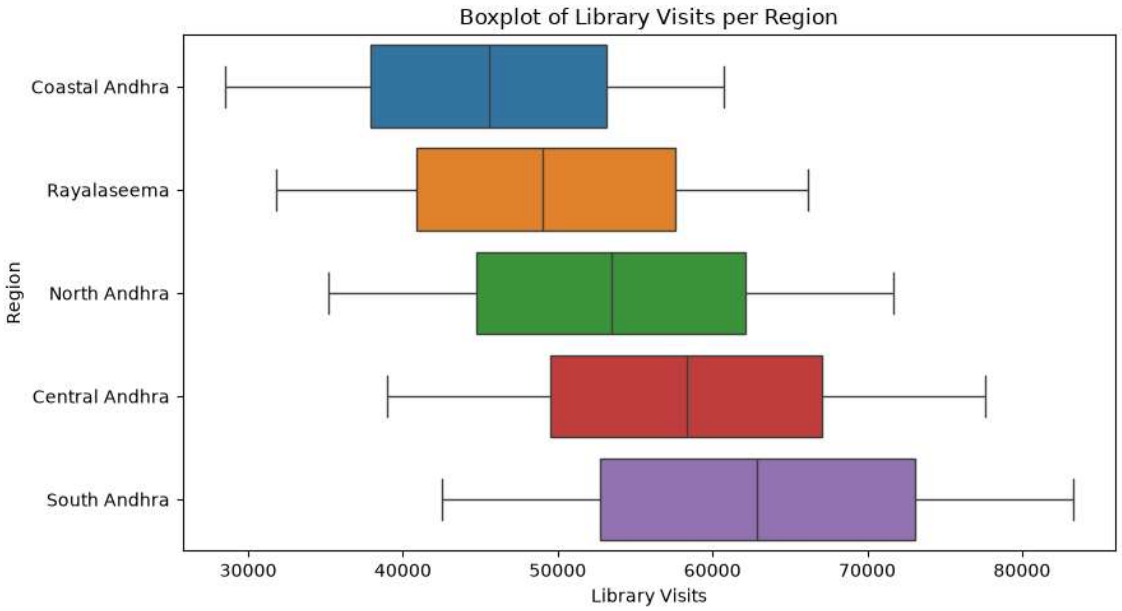


In [ ]:
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## Step 10 — Boxplot
```

In [10]:
```
plt.figure(figsize=(9, 5), dpi=100)

sns.boxplot(
    x="library_visits",
    y="state",
    data=df_241UDS0103,
    hue="state",
    legend=False
)

plt.title("Boxplot of Library Visits per Region")
plt.xlabel("Library Visits")
plt.ylabel("Region")
plt.tight_layout()
plt.show()
```



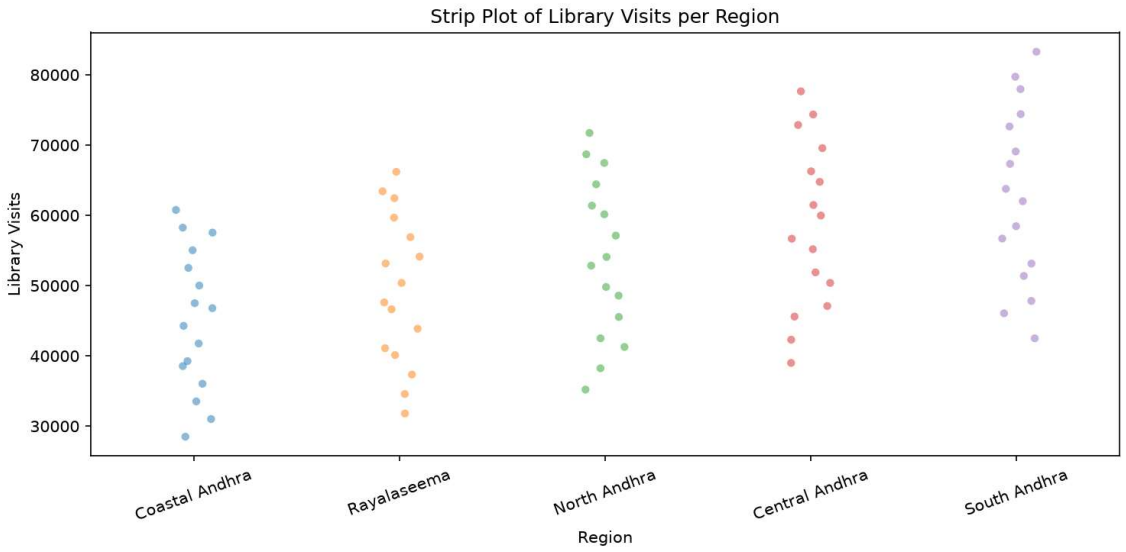
In [ ]:
```
## Step 11 — Swarm / Strip Plot
```

In [11]:

```
plt.figure(figsize=(10, 5), dpi=150)

sns.stripplot(
    x="state",
    y="library_visits",
    data=df_241UDS0103,
    hue="state",
    alpha=0.5,
    legend=False
)

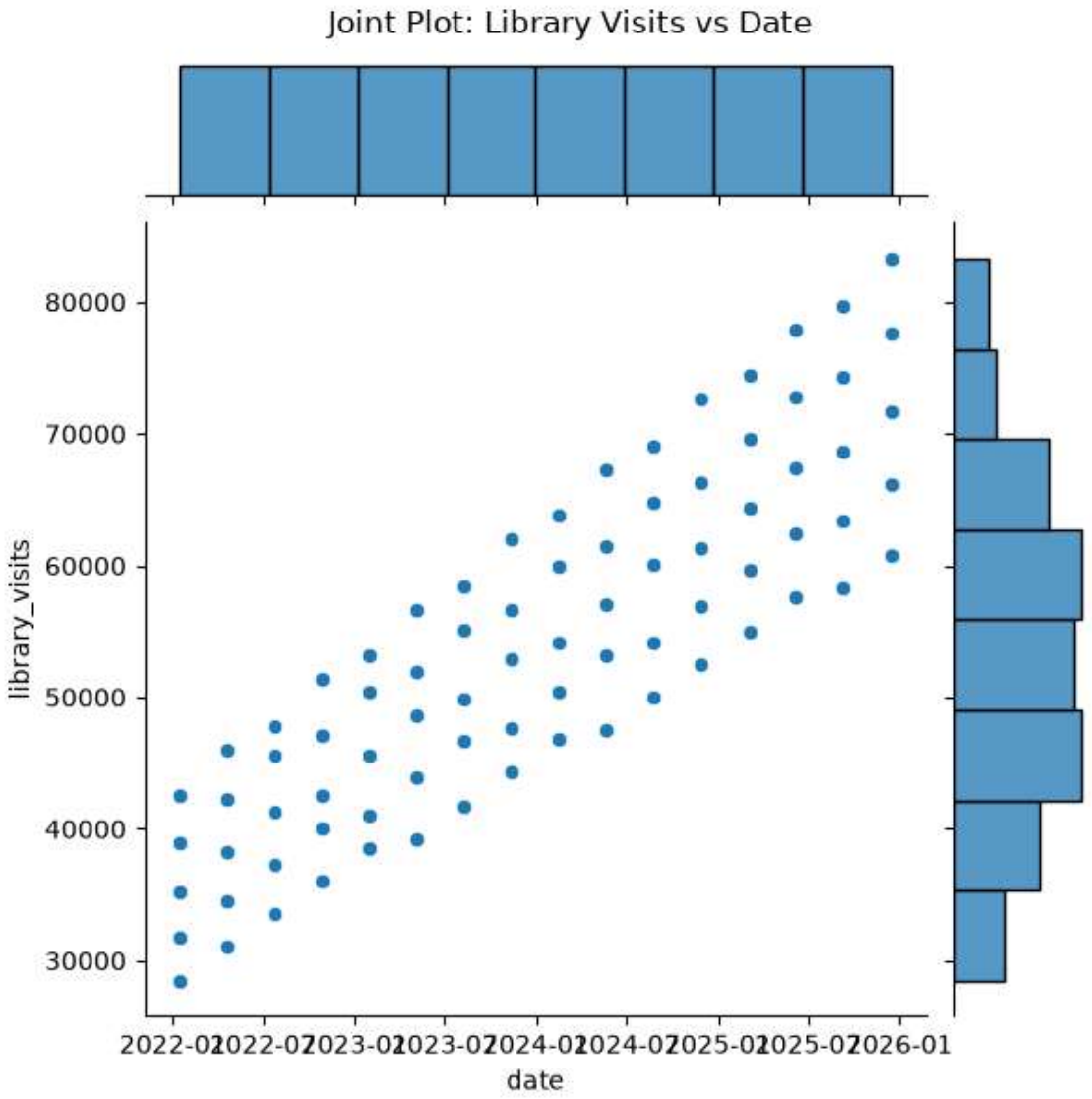
plt.title("Strip Plot of Library Visits per Region")
plt.xlabel("Region")
plt.ylabel("Library Visits")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [ ]:

```
## Step 12 – Joint Plot
```

In [12]:
```
sns.jointplot(
    x="date",
    y="library_visits",
    data=df_241UDS0103
)
plt.suptitle("Joint Plot: Library Visits vs Date", y=1.02)
plt.show()
```



In [ ]:
```
## Step 13 — Heat Map Preparation
```

In [13]:
```
# Average library visits per year per region
yearly_avg_241UDS0103 = df_241UDS0103.groupby(
    [ "year_241UDS0103", "state" ]
)["library_visits"].mean()

yearly_avg_241UDS0103
```

Out[13]:
```
year_241UDS0103  state
2022             Central Andhra    43495.5
                 Coastal Andhra    32266.5
                 North Andhra      39302.5
                 Rayalaseema       35959.5
                 South Andhra      46938.5
2023             Central Andhra    53533.5
                 Coastal Andhra    40960.5
                 North Andhra      49192.5
                 Rayalaseema       44801.5
                 South Andhra      57574.5
2024             Central Andhra    63121.5
                 Coastal Andhra    49204.5
                 North Andhra      58182.5
                 Rayalaseema       53643.5
                 South Andhra      68210.5
2025             Central Andhra    73609.5
                 Coastal Andhra    57898.5
                 North Andhra      68072.5
                 Rayalaseema       62935.5
                 South Andhra      78846.5
Name: library_visits, dtype: float64
```

In [ ]:
```
## Step 14 — Using unstack()
```

In [14]:
```
# Reshape the result: each row = year, each column = region
yearly_avg_241UDS0103 = df_241UDS0103.groupby(
    ["year_241UDS0103", "state"]
)["library_visits"].mean().unstack()

yearly_avg_241UDS0103.head()
```

Out[14]:

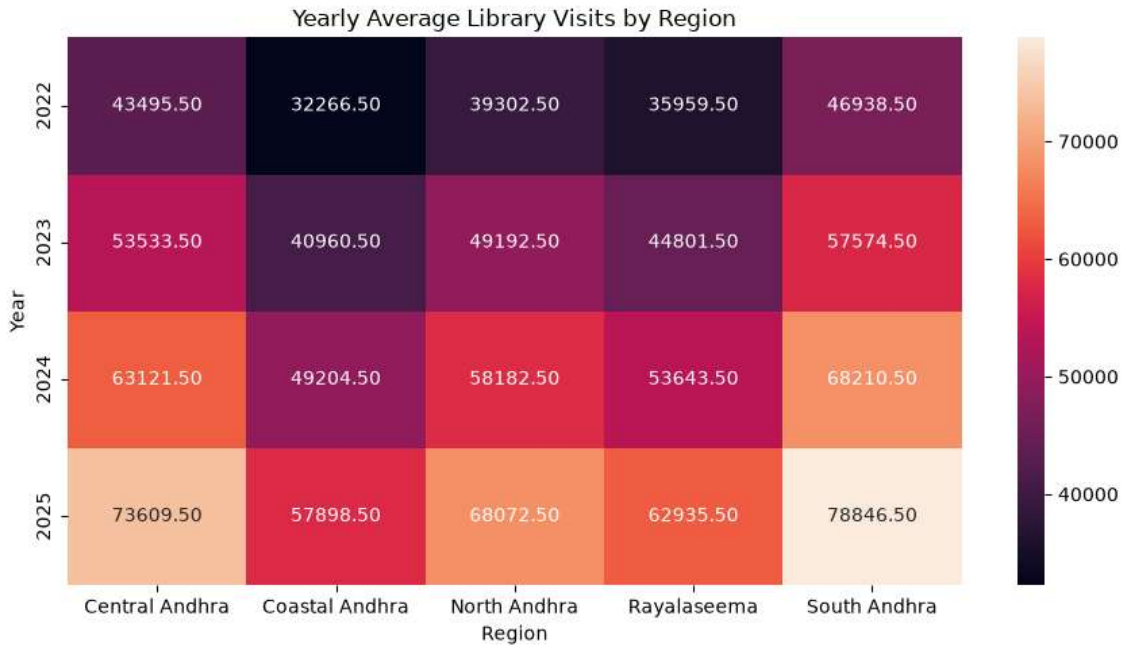
state	Central Andhra	Coastal Andhra	North Andhra	Rayalaseema	South Andhra
year_241UDS0103					
2022	43495.5	32266.5	39302.5	35959.5	46938.5
2023	53533.5	40960.5	49192.5	44801.5	57574.5
2024	63121.5	49204.5	58182.5	53643.5	68210.5
2025	73609.5	57898.5	68072.5	62935.5	78846.5

In [ ]:
```
## Step 15 — Heatmap
```

In [15]:
```
plt.figure(figsize=(9, 5), dpi=100)

sns.heatmap(
    yearly_avg_241UDS0103,
    fmt=".2f",
    annot=True
)

plt.title("Yearly Average Library Visits by Region")
plt.xlabel("Region")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```



In [ ]:
```
## RESULT

Various data-visualization techniques using the Seaborn library were
implemented to analyze and compare library visits across regions and years.

**Note:** The dataset is synthetic practice data. Region names are used as
categories to preserve the grouping structure required for the plots.
```